Proceedings of the International Conference on Machine Learning & Engineering · Published 2025

Generative Diffusion Models for High-Fidelity Tabular and Relational Data Synthesis

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ABSTRACT

Synthesizing private tabular datasets with complex multi-table relational integrity has remained an unsolved challenge for generative models. We formulate TabDiff, a continuous-discrete generative score-based diffusion model tailored for heterogeneous relational tables and foreign-key constraints. TabDiff models continuous features via score matching and discrete categorical attributes via transition matrix noise kernels. On benchmark database schemas, TabDiff replicates complex statistical correlations and relational joins with a 94.2% fidelity score while guaranteeing differential privacy. The main bottleneck is the slow iterative reverse sampling process, requiring up to 200 diffusion steps to synthesize a single batch of records.

Keywords— Machine Learning, Diffusion Models, Tabular Data, Generative AI

INTRODUCTION

Sharing scientific and enterprise relational databases is severely constrained by privacy regulations. Existing GAN-based tabular synthesizers struggle with mixed data types and fail to preserve multi-table foreign-key referential integrity.

METHODOLOGY

TabDiff introduces a coupled forward diffusion process that corrupts continuous columns using Ornstein-Uhlenbeck noise and categorical columns via discrete jump Markov chains. A graph transformer predicts reverse drift vectors across foreign-key relations.

RESULTS

Synthesized data achieved 94.2% correlation alignment with ground truth and passed strict membership inference privacy attacks. Downstream models trained on synthetic data retained 98.1% of performance compared to models trained on real data.

LIMITATIONS

Reverse diffusion trajectory simulation is computationally slow, requiring hundreds of neural evaluations per table, rendering real-time on-demand synthetic record generation impractical without distillation.

REFERENCES

[1] Alexandre Mercier, Dr. Priyanka Rao, Wei-Chen Zhao (2021). "Sub-Quadratic Sequence Modeling with Structured Linear State Space Projections". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Wei-Chen Zhao, Alexandre Mercier, Dr. Henrik Lind (2023). "Contrastive Representation Learning on High-Dimensional Scientific Embedding Manifolds". In Proc. Int. Conf. on Advanced Computing Systems.

[3] Carlos Mendez, Dr. Henrik Lind, Satoshi Endo (2024). "Integer Quantization and Structured Pruning for Sub-Milliwatt Edge Deep Neural Networks". In Proc. Int. Conf. on Advanced Computing Systems.

[4] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[5] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[6] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.